SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO1 - ASSESSMENT TOOL - 1

DESCRIPTIVE WRITTEN TEST

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Descriptive Written Test**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO1 Statement	*Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.* **(BL3)**		
Student Name	**V.Nitesh Kumar Reddy**	**Reg. No.**	**192511123**
Section/Batch	**Slot A / BE-CSE**	**Date**	**21-10-2026**

Code of Conduct:

I, V.Nitesh Kumar Reddy, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name :- V. Nitesh Kumar Reddy
Regester NO :- 19251123
Course Code :- CSA 3717
Course Name :- Software Testing

CO1_AT-1

18. Testing Principles and Boundary Testing to identify untested Edge Case.

Introduction :- Software testing is an important activity in software development that helps ensure a system works correctly, reliably and efficently before it is released to users. However, some systems fail after deployment because certain conditions, known as edge cases, were not properly tested during development

2. Why were Edge Cases Missed?
Several reasons can explain why edge cases were not identified during the testing phase.

2.1 Incomplete Test cases
Test cases may focus only on the most common operations performed by users. Developers and testers might verify that a system work with normal inputs but fail to test unsual or extreme inputs

2.2 Lack of Boundary value Testing

Many software defects occur at the boundaries of input ranges. If testers donot check the minimum and maximum values, the application may incorrectly process values at these limits.

For example, a system that accepts marks from 0 to 100 may incorrectly reject 100 or accept 101 due to a programming error.

3. How Boundary Testing Helps Identify Edge cases

Boundary value Analysis (BVA) is a software testing technique that checks values at the edges of an input range. It is based on the observation that defects frequently occur near the min and maxi permitted values.

Instead of testing only typical inputs, boundary testing examines values at and around the limits.

Example: Student mark validation.

Suppose a system accepts marks between 0 and 100, inclusive.

The following table illustrates boundary testing

Test case	Input marks	Expected result
Below minimum	-1	Reject
Minimum boundary	0	Accept
Just above minimum	1	Accept
Normal value	50	Accept
Just below maximum	99	Accept
Maximum boundary	100	Accept
Above maximum	101	Reject

⇒ Types of Boundary Testing

1. Minimum boundary testing :- Checks the smallest permitted value and values immediately below and above it.

2. Maximum boundary testing :- Checks ~~limits such~~ the largest permitted value and values immediately below and above it.

3. Boundary testing for sizes:- checks limits such as maximum file size, password length and the number of records allowed.

4. Boundary testing for time and dates:- concentrates testing checks situations such as midnight, the last day of a month, leap days, and the start or end of a subscription.

Boundary testing is useful because it concentrates testing efforts on areas where errors are more likely to occur. It can reveal incorrect comparison operators off-by-one errors, invalid range checks and improper input validation.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	